

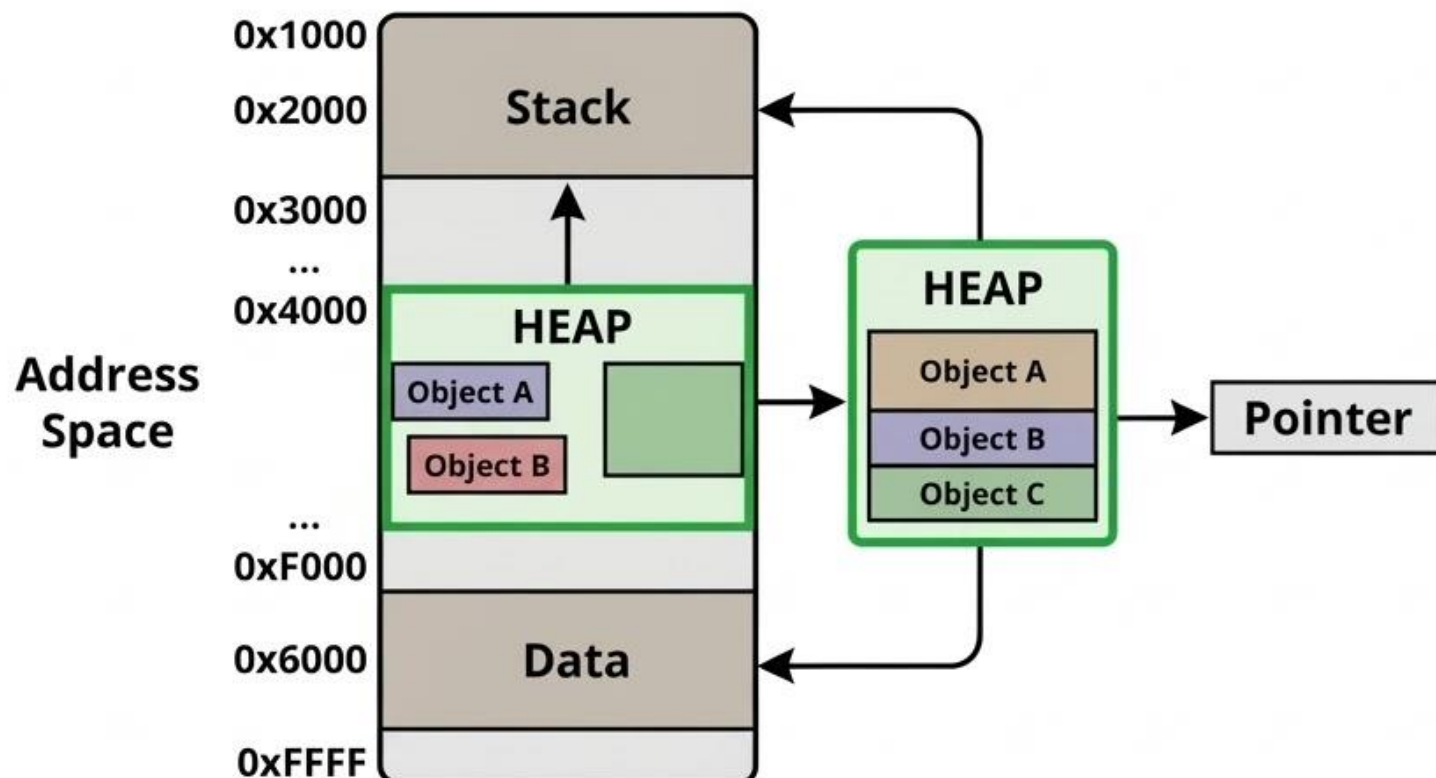
As a Database Storage

- 1. Rows in Pages**
- 2. Store data in disk**
- 3. Random order in Page**
- 4. Used in Database Table**
- 5. Insert: $O(1)$**
- 6. Read: $O(n) \rightarrow$ (without indexing)**



As a Memory Heap (RAM)

- 1. Unstructured Memory Region**
- 2. Store data in RAM**
- 3. Unordered / Managed via GC or Manual**
- 4. Runtime Memory Allocation**
- 5. Insert: $O(1)$ to $O(n)$**
- 6. Read: $O(1) \rightarrow$ (by memory address pointer)**



As a Data Structure

1. Binary Tree Structure

2. Store data in Memory

3. Min / Max ordered Tree

4. Used in Algorithms

5. Insert: $O(\log(n))$

6. Read: $O(n) \rightarrow$ (in worst case)

